

Air University



# Object Oriented Programming (Lab)

Year 2025

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Department of Cyber Security

USAMA ARSHAD | 247141

BS – Cyber Security – Fall – 24 (A)

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Lab 01

## Task 01:

## Code

```
1  #include <iostream>
2  #include <cstring>
3  using namespace std;
4
5  struct Address
6  {
7      int zip_code;
8      string address;
9  };
10 struct Student
11 {
12     int roll_no;
13     string name;
14     Address add;
15 };
16
17 void display (const Student &s)
18 {
19     cout << "\nRoll No: " << s.roll_no << endl;
20     cout << "Name: " << s.name << endl;
21     cout << "Address: " << s.add.address << endl;
22     cout << endl;
23 }
24
25 int main ()
26 {
27     Student s1;
28
29     cout << "Enter Roll No: ";
30     cin >> s1.roll_no;
31     cin.ignore();
32     cout << "Enter Name: ";
33     getline (cin, s1.name);
34
35     cout << "Enter Address: ";
36     getline (cin, s1.add.address);
37
38     cout << "Enter Zip Code: ";
39     cin >> s1.add.zip_code;
40     cin.ignore();
41
42     display (s1);
43
44     return 0;
45 }
```

## Output

```
Enter Roll No: 247141
Enter Name: Usama Arshad
Enter Address: Lalamusa
Enter Zip Code: 50200

Roll No: 247141
Name: Usama Arshad
Address: Lalamusa
```

## Task 02:

## Code

```
1  √ #include <iostream>
2  #include <cstring>
3  using namespace std;
4
5  √ struct Address
6  {
7      int zip_code;
8      string address;
9  };
10
11 √ struct Student
12 {
13     int roll_no;
14     string name;
15     Address add;
16 };
17
18 √ void display (const Student s[], int n)
19 {
20     √ for (int i = 0; i < n; i++)
21     {
22         cout << "\nStudents Details: " << endl;
23         cout << "Roll No: " << s[i].roll_no << endl;
24         cout << "Name: " << s[i].name << endl;
25         cout << "Address: " << s[i].add.address << endl;
26     }
27     cout << endl;
28 }
29
```

```
29
30  int main ()
31  {
32      const int max_students = 3;
33      Student s1[max_students];
34
35      for (int i = 0; i < max_students; i++)
36      {
37          cout << "Enter student " << i + 1 << " Roll No: ";
38          cin >> s1[i].roll_no;
39
40          cin.ignore();
41          cout << "Enter Student " << (i+1) << " Name: ";
42          getline (cin, s1[i].name);
43
44          cout << "Enter Student " << (i+1) << " Address: ";
45          getline (cin, s1[i].add.address);
46
47          cout << "Enter Student " << (i+1) << " Zip Code: ";
48          cin >> s1[i].add.zip_code;
49          cin.ignore();
50      }
51
52      display (s1, max_students);
53
54      return 0;
55  }
```

## Output

```
Enter student 1 Roll No: 247141
Enter Student 1 Name: Usama Arshad
Enter Student 1 Address: Lalamusa
Enter Student 1 Zip Code: 50200
Enter student 2 Roll No: 247171
Enter Student 2 Name: Asad Farooq
Enter Student 2 Address: Jhelum
Enter Student 2 Zip Code: 49600
Enter student 3 Roll No: 247127
Enter Student 3 Name: Hamza Ali
Enter Student 3 Address: Gujrat
Enter Student 3 Zip Code: 50600

Students Details:
Roll No: 247141
Name: Usama Arshad
Address: Lalamusa

Students Details:
Roll No: 247171
Name: Asad Farooq
Address: Jhelum

Students Details:
Roll No: 247127
Name: Hamza Ali
Address: Gujrat
```

**Task 03:****Code**

```
1  #include <iostream>
2  using namespace std;
3
4  class Date
5  {
6      private:
7          int day, month, years;
8
9      public:
10     void getData(int d, int m, int y)
11     {
12         day = d;
13         month = m;
14         years = y;
15     }
16     void displayDate()
17     {
18         cout << "\nCurrent Date: " << endl;
19         cout << day << "/" << month << "/" << years << endl;
20         cout << endl;
21     }
22 };
23
```

```
24  int main ()
25  {
26      int date, month, year;
27
28      cout << "\nEnter Day: ";
29      cin >> date;
30      while (date < 1 || date > 31)
31      {
32          cout << "Enter Valid Day: ";
33          cin >> date;
34      }
35
36      cout << "Enter Month: ";
37      cin >> month;
38      while (month < 1 || month > 12)
39      {
40          cout << "Enter Valid Month: ";
41          cin >> month;
42      }
```

```
43
44     cout << "Enter Year: ";
45     cin >> year;
46     while (year < 2025 || year > 2050)
47     {
48         cout << "Enter Valid Year: ";
49         cin >> year;
50     }
51
52     Date d1;
53     d1.getData(date, month, year);
54     d1.displayDate();
55
56     return 0;
57 }
```

### Output

```
Enter Day: 0
Enter Valid Day: 32
Enter Valid Day: 24
Enter Month: 0
Enter Valid Month: 13
Enter Valid Month: 02
Enter Year: 2024
Enter Valid Year: 2051
Enter Valid Year: 2025

Current Date:
24/2/2025
```

**THANK YOU FOR YOUR TIME!**

**THE END**